

*Part A***Question 1:****a)**

By Definition 1.9 Chapter 13 p63 and Theorem 1.22 Chapter 13 p 64;

i.

An example of a strictly monotonic decreasing sequence is;

$$(x_n) = 1 + \frac{1}{n}, \text{ for } n \geq 1$$

It is decreasing because;

$$\begin{aligned} x_{n+1} - x_n &= \left(1 + \frac{1}{n+1}\right) - \left(1 + \frac{1}{n}\right) \\ &= \frac{1}{n+1} - \frac{1}{n} &= \frac{n - (n+1)}{n(n+1)} \\ &= \frac{-1}{n(n+1)} < 0 \end{aligned}$$

for all $n \geq 1$

and,

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = 1 + 0 = 1$$

Thus by Theorem 1.22 Chapter 13 p 64, since (x_n) is strictly monotonic decreasing and bounded below by 1, it must converge to its greatest lower bound i.e. $\lim_{n \rightarrow \infty} x_n = 1$.

ii.

An example of a strictly monotonic increasing sequence is;

$$(y_n) = -1 - \frac{1}{n}, \text{ for } n \geq 1$$

It is increasing because;

$$\begin{aligned}
 y_{n+1} - y_n &= \left(-1 - \frac{1}{n+1}\right) - \left(-1 - \frac{1}{n}\right) \\
 &= -\frac{1}{n+1} + \frac{1}{n} = \frac{n - (n+1)}{n(n+1)} \\
 &= \frac{1}{n(n+1)} > 0
 \end{aligned}$$

for all $n \geq 1$

and,

$$\lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} \left(-1 - \frac{1}{n}\right) = -1 - 0 = -1$$

Thus by Theorem 1.22 Chapter 13 p 64, since (y_n) is strictly monotonic increasing and bounded above by -1, it must converge to its least upper bound i.e. $\lim_{n \rightarrow \infty} y_n = -1$.

b)

$$(z_n) = \begin{cases} x_n, & \text{if } n \text{ is odd} \\ y_n, & \text{if } n \text{ is even} \end{cases}$$

Where (x_n) is the strictly monotonic decreasing sequence defined in part (a)(i) and (y_n) is the strictly monotonic increasing sequence defined in part (a)(ii). We can see that the subsequence of odd terms (z_{2n-1}) is equivalent to (x_n) and the subsequence of even terms (z_{2n}) is equivalent to (y_n) . Thus we have that;

$$\lim_{n \rightarrow \infty} z_{2n-1} = \lim_{n \rightarrow \infty} x_n = 1$$

and

$$\lim_{n \rightarrow \infty} z_{2n} = \lim_{n \rightarrow \infty} y_n = -1$$

Since the two subsequences converge to different limits, the sequence (z_n) does not converge.

Question 2:

$$L = L_{a,\theta} = \{(x, a + x \tan \theta) : x \in \mathbb{R}\}$$

Where a is the y -intercept of the line and θ is the angle of inclination of the line to the positive x -axis.

$$X = L_{a,\theta} : a \in \mathbb{R} \text{ and } \theta \in (-\pi/2, \pi/2)$$

and

$$\Delta(L_{a,\theta}, L_{b,\phi}) = |a - b| + |\tan \theta - \tan \phi|$$

for $L_{a,\theta}, L_{b,\phi} \in X$

a)

The distance between $L_{0,\pi/4}$ and $L_{1,-\pi/4}$:

$$\begin{aligned} \Delta(L_{0,\pi/4}, L_{1,-\pi/4}) &= |0 - 1| + |\tan \pi/4 - \tan -\pi/4| \\ &= 1 + |1 - (-1)| \\ &= 1 + |2| \\ &= 1 + 2 = 3 \end{aligned}$$

b)

By Definition 2.1, HB Chapter 14 p72,

i.

To show Δ satisfies (M1) non-negativity, we need to show that for any

$$d^{(2)}(\mathbf{a}, \mathbf{b}) \geq 0$$

$$\Delta(L_{a,\theta}, L_{b,\phi}) = |a - b| + |\tan \theta - \tan \phi|$$

As $\mathbf{a}, \mathbf{b} \in \mathbb{R}$ and $\theta \in (-\pi/2, \pi/2)$, both $|a - b| \geq 0$ and $|\tan \theta - \tan \phi| \geq 0$

$$\geq 0 + 0$$

Thus, $\Delta(\mathbf{L}_{\mathbf{a},\theta}, \mathbf{L}_{\mathbf{b},\phi}) \geq 0$ and (M1) is satisfied.

ii.

To show Δ satisfies (M2) symmetry, we need to show that for any $d^{(2)}(\mathbf{a}, \mathbf{b}) = d^{(2)}(\mathbf{b}, \mathbf{a})$

$$\Delta(\mathbf{L}_{\mathbf{a},\theta}, \mathbf{L}_{\mathbf{b},\phi}) = |a - b| + |\tan \theta - \tan \phi|$$

By the properties of absolute values, HB p8, $|a - b| = |b - a|$ and $|\tan \theta - \tan \phi| = |\tan \phi - \tan \theta|$

$$= |b - a| + |\tan \phi - \tan \theta|$$

$$= \Delta(\mathbf{L}_{\mathbf{b},\phi}, \mathbf{L}_{\mathbf{a},\theta})$$

Thus, $\Delta(\mathbf{L}_{\mathbf{a},\theta}, \mathbf{L}_{\mathbf{b},\phi}) = \Delta(\mathbf{L}_{\mathbf{b},\phi}, \mathbf{L}_{\mathbf{a},\theta})$ and (M2) is satisfied.

iii.

To show Δ satisfies (M3) triangle inequality, we need to show that for any $d^{(2)}(\mathbf{a}, \mathbf{c}) \leq d^{(2)}(\mathbf{a}, \mathbf{b}) + d^{(2)}(\mathbf{b}, \mathbf{c})$

$$\Delta(\mathbf{L}_{\mathbf{a},\theta}, \mathbf{L}_{\mathbf{c},\psi}) = |a - c| + |\tan \theta - \tan \psi|$$

By the triangle inequality for real numbers, HB p9,

$$\leq (|a - b| + |b - c|) + (|\tan \theta - \tan \phi| + |\tan \phi - \tan \psi|)$$

$$= |a - b| + |\tan \theta - \tan \phi| + |b - c| + |\tan \phi - \tan \psi|$$

$$= \Delta(\mathbf{L}_{\mathbf{a},\theta}, \mathbf{L}_{\mathbf{b},\phi}) + \Delta(\mathbf{L}_{\mathbf{b},\phi}, \mathbf{L}_{\mathbf{c},\psi})$$

Thus, $\Delta(\mathbf{L}_{\mathbf{a},\theta}, \mathbf{L}_{\mathbf{c},\psi}) \leq \Delta(\mathbf{L}_{\mathbf{a},\theta}, \mathbf{L}_{\mathbf{b},\phi}) + \Delta(\mathbf{L}_{\mathbf{b},\phi}, \mathbf{L}_{\mathbf{c},\psi})$ and (M3) is satisfied.

c)

$$\mathbf{B}_{\Delta}(\mathbf{L}_{0,0}, 1)$$

By Definition 2.6, HB Chapter 14 p72,

$$\begin{aligned}\mathbf{B}_{\Delta}(\mathbf{L}_{0,0}, 1) &= \{\mathbf{L}_{b,\theta} \in \mathbf{X} : \Delta(\mathbf{L}_{0,0}, \mathbf{L}_{b,\theta}) < 1\} \\ &= \{\mathbf{L}_{b,\theta} \in \mathbf{X} : |0 - b| + |\tan 0 - \tan \theta| < 1\} \\ &= \{\mathbf{L}_{b,\theta} \in \mathbf{X} : |b| + |0 - \tan \theta| < 1\} \\ &= \{\mathbf{L}_{b,\theta} \in \mathbf{X} : |b| + |\tan \theta| < 1\}\end{aligned}$$

Applying the triangle inequality in $\mathbb{R}$ to each absolute value term,

$$\{\mathbf{L}_{b,\phi} \in \mathbf{X} : |b| + |\tan \phi| < 1\}$$

d)

The sequence of lines;

$$(\mathbf{L}_n)_{n=1}^{\infty} \text{ given by } \mathbf{L}_n = \mathbf{L}_{-1/n, \pi/2 - 1/n}$$

is not Δ -convergent because, by Definition 3.2 HB Chapter 14 p73;

Using $\mathbf{L}_{b,\phi} \in \mathbf{X}$ Then

$$\Delta(\mathbf{L}_n, \mathbf{L}_{b,\phi}) = \left| -\frac{1}{n} - b \right| + \left| \tan\left(\frac{\pi}{2} - \frac{1}{n}\right) - \tan \phi \right|$$

As $n \rightarrow \infty$, we have $-1/n \rightarrow 0$, so

$$\left| -\frac{1}{n} - b \right| \rightarrow |b|$$

Also, as $n \rightarrow \infty$, we have

$$\tan\left(\frac{\pi}{2} - \frac{1}{n}\right) \rightarrow +\infty$$

and since $\tan \phi$ is finite for all $\phi \in (-\pi/2, \pi/2)$,

$$\left| \tan\left(\frac{\pi}{2} - \frac{1}{n}\right) - \tan \phi \right| \rightarrow \infty.$$

Hence $\Delta(\mathbf{L}_n, \mathbf{L}_{b,\phi}) \rightarrow \infty$, and so $(\mathbf{L}_n)$ is not Δ -convergent.

Question 3:

$$f_n(x) = \begin{cases} \sin(\pi x), & \text{if } 2n \leq x < (2n+1), \\ 0, & \text{otherwise.} \end{cases}$$

a)**b)**

To test for pointwise limits we can take $x = 2.5$ as a constant value.

When $n = 1$ the interval is $[2, 3)$

$$f_1(2.5) = \sin(2.5\pi) = 1$$

When $n = 2$ the interval is $[4, 5)$

$$f_2(2.5) = 0$$

When $n = 3$ the interval is $[6, 7)$

$$f_3(2.5) = 0$$

$$\vdots$$

When $n = k$ the interval is $[2k, 2k+1)$

$$f_k(2.5) = 0$$

$$\forall x \in \mathbb{R}, \lim_{n \rightarrow \infty} = 0$$

Therefore $f_n \rightarrow f$ pointwise as $n \rightarrow \infty$

c)

Using Definition 4.4 Chapter 15 p 78 we can test for uniform convergence.

Assume $\epsilon = \frac{1}{2}$

$$|f_n(x) - f(x)| < \epsilon$$

$$|\sin(\pi x) - 0| < \frac{1}{2}$$

$$|\sin(\pi x)| < \frac{1}{2}$$

This is true for some values of x but not for all values of x in the domain of $f_n(x)$.

Therefore f_n does not converge uniformly to f as $n \rightarrow \infty$